CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT)

SEM - II University Examination, November -2012

MA - 102 Engineering Mathematics-II

Date: 14.12.12, Friday **Time: 1.30 p. m to 4.30 p. m.** **Maximum Marks: 70**

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 (a) Solve : $(x^2-4xy-2y^2)\,dx+(y^2-4xy-2x^2)\,dy=0$. **[04]**

(b) Solve : $p^2+q^2=1$. **[03]**

Q-2

(a) Solve by the method of variation of parameters $\dfrac{d^2y}{dx^2}+16y=\sec 4x$. **[05]**

(b) Solve: (i) $y''+9y'+20y=0$ (ii) $(4D^2+20D+25)y=0$. **[05]**

(c) Solve the initial-value problem: $y''+2y'+y=e^{-2x}$, $y(0)=-1, y'(0)=1$. **[04]**

OR

Q-2 (a) Solve: $\dfrac{dy}{dx}+y\tan x=\sin 2x$, $y(0)=1$. **[05]**

(b) Solve: $y'''+3y''+3y'+y=8e^x+x+3$. **[05]**

(c) Solve the initial-value problem: $x^2y''+xy'+y=x^2+1$. **[04]**

Q-3 (a) Solve $x\left(z^2-y^2\right)p+y\left(x^2-z^2\right)q=z\left(y^2-x^2\right)$.. **[05]**

(b) Solve (i) $p(1+q)=qz$ (ii) $z-px-qy=2\sqrt{pq}$ **[05]**

(c) Solve : $p-x^2=q+y^2$. **[04]**

OR

Q-3 (a) Solve $(mz-ny)\dfrac{\partial z}{\partial x}+(nx-lz)\dfrac{\partial z}{\partial y}=ly-mx$. **[05]**

(b) Solve (i) $p^2-q^2=x-y$ (ii) $z=p^2+q^2$. **[05]**

(c) Solve: $p+q+pq=0$. **[04]**

SECTION-II

Q-4 (a) Prove that $A=\frac{1}{2}\begin{bmatrix}1+i & -1+i\\ 1+i & 1-i\end{bmatrix}$ is a Unitary matrix. **[03]**

(b) Find the area between the parabola $y^2=4ax$ and $x^2=4ay$. **[04]**

Q-5 (a) Find the eigenvalues and corresponding eigenvectors for the **[05]**

Matrix $A=\begin{bmatrix}3 & 1 & 4\\ 0 & 2 & 6\\ 0 & 0 & 5\end{bmatrix}$

(b) Evalate $\iint r^3\,dr\,d\theta$ over the area included between the circles $r=2\sin\theta$ and $r=4\sin\theta$. **[05]**

(c) Prove that $J_{-\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\cos x$. **[04]**

OR

Q-5

(a) Find the characteristic equations of the matrix $A=\begin{bmatrix}2 & -1 & 1\\ -1 & 2 & -1\\ 1 & -1 & 2\end{bmatrix}$ and evaluate the **[05]**

matrix equations $A^6-6A^5+9A^4-2A^3-12A^2+23A-9I$ and hence obtain A^{-1}.

(b) Change the order of integration in the integral $\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}\,dy\,dx$ and evaluate it. **[05]**

(c) Prove that $\int_0^1\int_0^x\int_0^{\sqrt{x+y}} z\,dx\,dy\,dz=\frac{1}{4}$. **[04]**

OR

Q-6

(a) If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, compute $A^2-4A-5I$ and A^{-1}. **[05]**

(b) Express $f(x)=4x^3-3x^2+2x+1$ in terms of Legendre polynomials. **[05]**

(c) Evaluate $\int_0^{\infty} x^2 e^{-x^4}\,dx$. **[04]**

OR

Q-6 (a)State Orthogonality property of Legendre polynomial $P_n(x)$ and

Also, find (i) $\int_{-1}^{1} P_3(x)P_4(x)\,dx=$__________ (ii) $\int_{-1}^{1} P_6^2(x)\,dx=$__________. **[05]**

(b) Adopting standard notations, Prove that $\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)$. **[05]**

(c) Evaluate $\iint_R dA$; R is the region bounded by $x=0, x=1, y=x^2, y=2-x$. **[04]**